

## GO\_Biological\_Process\_2023 Top pathways by non-permutation

| Geneset                                   | stat        | num.genes | pval      | p.adj     | gene.vals                                                                 |
|-------------------------------------------|-------------|-----------|-----------|-----------|---------------------------------------------------------------------------|
| Alpha-Amino Acid Biosynthetic Process (G  | 0.26902178  | 10        | 3.222e-03 | 9.205e-01 | CAD:346 BCAT2:565 SDS:598 SDSL:604 MTHFD1:847 ILVB1:1549                  |
| Bile Acid And Bile Salt Transport (GO:00  | 0.20679031  | 16        | 4.191e-03 | 9.205e-01 | ABCC11:4 SLC10A6:86 SLC01A2:455 SLC27A5:1052 SLC01B1:1485 SLC10A2:1495    |
| Cellular Response To Acetylcholine (GO:1  | -0.30102484 | 10        | 9.804e-04 | 9.205e-01 | CHRN2:60 CHRM3:241 CHRNA3:347 LVH3:375 CHRND:603 LYPD1:2204               |
| Chromatin Remodeling (GO:0006338)         | 0.06047363  | 200       | 3.297e-03 | 9.205e-01 | HIRA:7 CHD8:27 GATAD2A:43 DAXX:80 RNF168:116 DDX12:145                    |
| Cilium-Dependent Cell Motility (GO:00602  | -0.20843465 | 17        | 2.931e-03 | 9.205e-01 | DNAH2:9 TEK1:157 DNAH17:186 TEK2:1032 GAS8:1325 TEK14:1573                |
| Glycerophospholipid Metabolic Process (G  | 0.22247174  | 56        | 1.501e-03 | 9.205e-01 | PLA2G2D:205 ABHD3:368 PLA2G7:432 PIGG:586 SMPD4:707 PIGV:769              |
| Mitochondrial Cytochrome C Oxidase Assem  | -0.21434926 | 15        | 4.038e-03 | 9.205e-01 | FASTKD3:648 TIMM21:735 COA1:1637 COX16:1767 SURF1:1958 COA8:2061          |
| Muscle Contraction (GO:0006936)           | -0.09908076 | 87        | 1.422e-03 | 9.205e-01 | MYH13:18 TACR2:22 CHRN2:60 ARHGEF11:147 MYH8:352 CHRN1:383                |
| Negative Regulation Of CD8-positive, Alp  | -0.34738851 | 6         | 3.210e-03 | 9.205e-01 | ZBTB78:227 SOCS1:511 HFE:1822 VSIR:1946 SLC4A2:2454 DAPL1:7900            |
| Neuropeptide Signaling Pathway (GO:00072  | -0.11308371 | 63        | 1.927e-03 | 9.205e-01 | NPBW2:75 SSTR4:256 CYSLTR2:353 OPRK1:396 RXFP3:397 GALLR1:440             |
| Nuclear Pore Organization (GO:0006999)    | 0.21692473  | 14        | 4.955e-03 | 9.205e-01 | NUP98:502 NDC1:514 NUP133:946 NUP107:1042 NUP205:1069 NUP54:1249          |
| Positive Regulation Of Cell Cycle Process | 0.08631847  | 105       | 2.316e-03 | 9.205e-01 | KIF14:54 ZNF16:125 EGF:181 KIF20B:276 STL:291 SMC2:311                    |
| Positive Regulation Of Chromosome Segreg  | 0.21401943  | 17        | 2.254e-03 | 9.205e-01 | SMC2:311 NUMA1:606 CDC6:885 NCAPH:1298 RCC2:1413 SMC4:1577                |
| Positive Regulation Of Histone Deacetyla  | 0.26303025  | 12        | 1.607e-03 | 9.205e-01 | SMARCA5:399 PRKD2:1177 VEGFA:1244 SIRT1:1319 BAZ2A:1376 SIRT6:1400        |
| Positive Regulation Of Protein Deacetyla  | 0.24504001  | 13        | 2.222e-03 | 9.205e-01 | SMARCA5:399 FNTA:887 IFNG:1099 SIRT1:1319 BAZ2A:1376 NIPBL:2199           |
| Positive Regulation Of Transcription By   | 0.29906611  | 12        | 3.346e-04 | 9.205e-01 | CHD8:27 DDX21:145 ERCC26:204 SMARCA5:399 ELL:471 ZC3H8:725                |
| Positive Regulation Of Tyrosine Phosphor  | 0.11929692  | 49        | 3.890e-03 | 9.205e-01 | TNF:259 IL23R:345 IGF1:669 PARP9:686 CNTF:754 CSF2:789                    |
| Preassembly Of GPI Anchor In ER Membrane  | 0.30978755  | 7         | 4.535e-03 | 9.205e-01 | PIGG:586 PIGV:769 PIGB:788 PIGN:932 PIGW:2007 PIGL:8068                   |
| Regulation Of Rho Protein Signal Transdu  | -0.10368282 | 68        | 3.141e-03 | 9.205e-01 | EPS8L3:57 RIPOR2:104 KANK1:117 ARHGEF2:221 GPR35:441 KANK2:601            |
| Regulation Of Cell Cycle (GO:0051726)     | -0.05211947 | 260       | 3.970e-03 | 9.205e-01 | BIRC3:45 RNF167:132 BIRC7:136 TBGR4:169 FIGNL1:292 ZBTB17:380             |
| Regulation Of Chromosome Separation (GO:  | 0.27222758  | 9         | 4.685e-03 | 9.205e-01 | SMC2:311 NUMA1:606 NCAPH:1298 SMC4:1577 NCAPG2:1675 NCAPD2:2960           |
| Regulation Of Interleukin-13 Production   | 0.27427262  | 9         | 4.383e-03 | 9.205e-01 | IRF4:529 SPHK2:530 IL4:952 TNFRSF21:1288 TSLP:2022 LEF1:2815              |
| Regulation Of Neuron Projection Developm  | -0.06774229 | 165       | 2.752e-03 | 9.205e-01 | MINAR1:7 CAMK2B:66 KANK1:117 ITGA6:199 ROR2:308 TENM3:434                 |
| Regulation Of Transcription By RNA Polym  | 0.20191780  | 22        | 2.046e-03 | 9.205e-01 | CHD8:27 DDX21:145 ERCC6:204 DHX36:258 SMARCA5:399 ELL:471                 |
| Serine Family Amino Acid Biosynthetic Pr  | 0.21722738  | 9         | 4.839e-03 | 9.205e-01 | CTH:608 SERINC3:745 MTHFD1:847 SRR:1930 PSHP:2220 SERINC5:2248            |
| sRNA Transcription By RNA Polymerase II   | 0.31382801  | 7         | 4.036e-03 | 9.205e-01 | SNAPC1:277 ELL:471 ZC3H8:725 ICE2:762 SNAPC5:2371 ELL:471                 |
| snRNA Transcription By RNA Polymerase I   | 0.43514501  | 4         | 2.577e-03 | 9.205e-01 | SNAPC1:277 ZC3H8:725 ICE2:762 SNAPC5:2371 NA NA                           |
| Steroid Hormone Biosynthetic Process (GO  | 0.21240953  | 16        | 3.269e-03 | 9.205e-01 | HSD17B6:146 CYP17A1:593 HSD17B3:600 HSD17B2:613 HSD17B11:893 CYP11A1:1087 |
| Steroid Metabolic Process (GO:0008202)    | 0.11549574  | 61        | 1.254e-03 | 9.205e-01 | HSD17B6:146 CYP51A1:314 HSD11B1:413 CYP17A1:593 HSD17B3:600 HSD17B2:613   |
| Glut Movement Involved In Cell Motilit    | -0.30353125 | 7         | 5.419e-03 | 9.450e-01 | TEKT1:157 TEKT2:1023 GAS8:1325 TEKT4:208 TEKT5:2367 RSPH9:7900            |
| Mitotic G1 DNA Damage Checkpoint Signali  | -0.15646549 | 26        | 5.767e-03 | 9.450e-01 | SDE2:264 MUC1:377 GML:389 PML:593 PLK3:753 GTS1:952                       |
| Negative Regulation Of Endothelial Cell   | 0.15949878  | 25        | 5.788e-03 | 9.450e-01 | NFE2L2:160 KRIT1:942 IL4:952 ICAM1:1194 FGG:1294 TEK:1440                 |
| Neutral Amino Acid Transport (GO:0015804  | 0.13712107  | 34        | 5.678e-03 | 9.450e-01 | SLC43A1:32 SLC6A6:143 SLC7A6:157 SLC36A2:219 SLC36A1:338 SLC6A9:362       |
| DNA Integrity Checkpoint Signaling (GO:0  | 0.11558151  | 36        | 1.645e-02 | 9.862e-01 | TP53BP1:94 ERCC26:204 CDC45:603 PARP9:686 CLSPN:864 CDC6:885              |
| DNA Replication Initiation (GO:0006270)   | 0.16029282  | 20        | 2.130e-02 | 9.862e-01 | ORC1:65 ORC3:597 CDC45:603 ORC2:1104 LRWD1:1396 PRIM2:1588                |
| L-phenylalanine Catabolic Process (GO:00  | 0.31263606  | 5         | 1.031e-02 | 9.862e-01 | TAT:592 HGD:715 QDDR:1714 IL4I1:2352 GSTZ1:8068 NA                        |
| L-phenylalanine Metabolic Process (GO:00  | 0.31263606  | 5         | 1.031e-02 | 9.862e-01 | TAT:592 HGD:715 QDDR:1714 IL4I1:2352 GSTZ1:8068 NA                        |
| L-serine Metabolic Process (GO:0006563)   | 0.24584037  | 10        | 7.106e-03 | 9.862e-01 | SDS:598 SDSL:604 SERINC3:745 CBS:1692 SRR:1930 PSHP:2220                  |
| Anterograde Trans-Synaptic Signaling (GO  | -0.05418828 | 180       | 1.242e-02 | 9.862e-01 | MINK1:56 CHRN2:60 HTR3B:177 CHRM3:241 CHRNA3:347 CHRN1:383                |
| Aromatic Compound Catabolic Process (GO:  | 0.26068214  | 7         | 1.692e-02 | 9.862e-01 | CYP2F1:30 TET1:719 PON3:900 PDXP:1354 PON1:8068 PON2:8068                 |

## EnrichmentHsSymbolsFile2 Top pathways by non-permutation

| Geneset                                    | stat        | num.genes | pval      | p.adj     | gene.vals                                                              |
|--------------------------------------------|-------------|-----------|-----------|-----------|------------------------------------------------------------------------|
| NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO   | -0.17013844 | 127       | 3.725e-11 | 2.417e-07 | LAMA5:61 HELZ2:64 NPBWR2:75 BIRC7:136 GAT5:200 ABHD16B:224             |
| HAMAL_APOPTOSIS_VIA_TRAIL_UP               | 0.07614772  | 591       | 3.180e-10 | 1.032e-06 | TASOR:14 SPG11:25 SMCHD1:35 CEP192:48 KIF14:54 NEK1:64                 |
| DACOSTA_UV_RESPONSE_VIA_ERCC3_DN           | 0.06143360  | 809       | 3.778e-09 | 8.171e-06 | TMEM131L:6 HIRA:7 MAN2A1:13 TASOR:14 UBR2:18 SPG11:25                  |
| NIKOLSKY_BREAST_CANCER_IQ21_AMPLICON       | -0.28843801 | 33        | 9.833e-09 | 1.595e-05 | DCST1:51 ADAM15:99 MTX1:134 PBXIP1:142 RUSC1:181 ZBTB7B:227            |
| DACOSTA_UV_RESPONSE_VIA_ERCC3_COMMON_DN    | 0.07693100  | 441       | 3.458e-08 | 4.487e-05 | HIRA:7 MAN2A1:13 TASOR:14 UBR2:18 SMCHD1:35 ZNF292:40                  |
| NIKOLSKY_BREAST_CANCER_16Q24_AMPLICON      | -0.21378117 | 46        | 5.299e-07 | 5.730e-04 | PIEZO1:11 CPNE7:192 ZC3H18:229 ANKRD11:334 PABPN1L:456 C8FA2T3:522     |
| NIKOLSKY_BREAST_CANCER_21Q22_AMPLICON      | -0.37576264 | 14        | 1.127e-06 | 1.044e-03 | COL6A1:73 COL6A2:87 SPATC1L:165 PCNT:236 DIP2A:606 COL18A1:808         |
| MARTENS_TRETINOIN_RESPONSE_UP              | -0.05423703 | 675       | 1.803e-06 | 1.462e-03 | B3GN76:19 SMTNL2:29 MMCL1:30 CARD14:41 C1QTNF8:43 COL6A1:73            |
| NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON      | -0.34442946 | 15        | 3.863e-06 | 2.703e-03 | SFSWAP:17 GALTNP:616 NOC4L:631 PUS1:636 GOLGA3:676 EPO40:682           |
| REACTOME_SIGNALING_BY_GPCR                 | -0.05401748 | 623       | 4.737e-06 | 2.703e-03 | TACR2:22 GPR132:47 CAMK2B:66 NPBWR2:75 BDKRB1:86 ARHGEF11:147          |
| REACTOME_CLASS_A_1_RHODOPSIN_LIKE_RECEPTOR | -0.07733324 | 296       | 4.999e-06 | 2.703e-03 | TACR2:22 GPR132:47 NPBWR2:75 BDKRB1:86 GPR68:150 CHRM3:241             |
| DAZARD_RESPONSE_TO_UV_NHEK_DN              | 0.07996617  | 280       | 4.858e-06 | 2.703e-03 | TMEM131L:6 WDFY3:37 KIF14:54 AKAP9:82 DOP1A:95 DST:106                 |
| WP_GPCRS_CLASS_A_RHODOPSIN_LIKE            | -0.08442194 | 211       | 2.453e-05 | 1.224e-02 | NPBW2:75 BDKRB1:86 GPR68:150 OR6B1:207 CHRM3:241 SSTR4:256             |
| STARK_HYPOCAMPUS_22Q11_DELETION_DN         | 0.25685484  | 22        | 3.042e-05 | 1.410e-02 | HIRA:7 PRODH:234 TANGO2:283 DGCRC6:343 DGCRC2:538 COMT:948             |
| HSIAO_LIVER_SPECIFIC_GENES                 | 0.08490669  | 199       | 3.749e-05 | 1.621e-02 | GSTO1:20 F11:57 SERPIND1:67 ALDH1A1:85 ALB:142 MGST1:216               |
| PEREZ_TP63_TARGETS                         | -0.06475642 | 313       | 8.548e-05 | 3.262e-02 | CEMP1:6 NLRP1:49 KLIK15:67 SEMA4D:155 APC2:212 EVPL:226                |
| SCHLOSSER_SERUM_RESPONSE_DN                | 0.04662107  | 618       | 8.344e-05 | 3.262e-02 | TASOR:14 UBR2:18 HEXB:21 ICAM2:28 ASMTL:29 PTPRC:47                    |
| REACTOME_TRANSPORT_OF_INORGANIC_CATIONS    | 0.10797444  | 101       | 1.792e-04 | 6.459e-02 | SLC38A1:32 SLC6A6:143 SLC7A6:157 SLC36A2:219 SLC26A3:273 SLC36A1:338   |
| PYEON_CANCER_HEAD_AND_NECK_VS_CERVICAL_U   | 0.08397830  | 166       | 1.930e-04 | 6.591e-02 | NR1D2:91 WDR76:134 SLF1:196 NUP210:202 HLF:206 TUG1:243                |
| WP_2Q112_COPY_NUMBER_VARIATION_SYNDROME    | 0.18366797  | 34        | 2.108e-04 | 6.837e-02 | ARID5A:98 CNMNA:178 MMS1:290 CNMNA:170 KANSL3:474 COL2A1:503           |
| RODRIGUES_THYROID_CARCINOMA_POORLY_DIFFE   | 0.04466657  | 586       | 2.384e-04 | 7.366e-02 | KIF14:54 WDR76:134 APC:147 AGGF1:167 SLF1:196 HLF:206                  |
| REACTOME_GPCR_LIGAND_BINDING               | -0.05305992 | 398       | 2.553e-04 | 7.370e-02 | TACR2:22 GPR132:47 NPBWR2:75 BDKRB1:86 GPR68:150 CHRM3:241             |
| DUTERTRE_ESTRADIOL_RESPONSE_24HR_UP        | 0.06065809  | 308       | 2.613e-04 | 7.370e-02 | DEPTOR:88 WDR76:134 TET2:179 IMPA2:208 SPDL1:227 KNCV1:5:242           |
| ZHANG_BREAST_CANCER_PROGENITORS_UP         | 0.05210130  | 408       | 3.214e-04 | 8.687e-02 | PROM1:1 SMCHD1:35 CEP192:48 ANAPC1:62 NEK1:64 AKAP9:82                 |
| WP_PEPTIDE_GPCRS                           | -0.12205469 | 70        | 4.165e-04 | 1.055e-01 | TACR2:22 BDKRB1:86 SSTR4:256 OPRK1:396 GALLR1:440 FSHR:490             |
| WP_22Q112_COPY_NUMBER_VARIATION_SYNDROME   | 0.10012894  | 104       | 4.230e-04 | 1.055e-01 | HIRA:7 CLTC1:19 PAKA:36 SERPIND1:67 SLC7A4:120 DEPDC5:136              |
| DODD_NASOPHARYNGEAL_CARCINOMA_DN           | 0.03016670  | 1204      | 4.957e-04 | 1.191e-01 | GSTO1:20 CEP192:48 KIF14:54 ANAPC1:62 EMILIN2:66 QSER1:127             |
| FISCHER_DREAM_TARGETS                      | 0.03346534  | 865       | 9.215e-04 | 2.135e-01 | MTBP:5 SMCHD1:35 GATAD2A:43 FNBP4:44 CEP192:48 KIF14:54                |
| REACTOME_PEPTIDE_LIGAND_BINDING_RECEPTOR   | -0.07399589 | 168       | 9.673e-04 | 2.164e-01 | TACR2:22 NPBWR2:75 BDKRB1:86 SSTR4:256 C3:296 NPW:340                  |
| ASGHARZADEH_NEUROBLASTOMA_POOR_SURVIVAL    | -0.15004354 | 39        | 1.188e-03 | 2.569e-01 | ADCY1:495 C1orf35:576 GATAD2B:775 HAP1:901 H2AZ2:999 TMOD2:1054        |
| CAIRO_HEPATOBLASTOMA_DN                    | 0.06121765  | 234       | 1.285e-03 | 2.688e-01 | TMEM131L:6 F11:57 OGDHL:93 CA2:140 HSD17B6:146 LEPR:190                |
| BENPORATH_ES_WITH_H3K27ME3                 | -0.03042522 | 976       | 1.434e-03 | 2.745e-01 | CEMP1:6 PLEC:12 BNC1:28 ANKRD35:31 CAMK2B:66 LONRF3:85                 |
| DAZARD_UV_RESPONSE_CLUSTER_G6              | 0.07923190  | 137       | 1.382e-03 | 2.745e-01 | KIF14:54 DOP1A:95 TUT2:156 UBR5:166 ZNF451:187 OTUD4:201               |
| TOYOTA_TARGETS_OF_MIR34B_AND_MIR34C        | 0.04654378  | 401       | 1.439e-03 | 2.745e-01 | PPP4R1:10 KIF14:54 F4R2:118 DDX21:145 NUP210:202 RAP1GAP:232           |
| REACTOME_ERYTHROCYTES_TAKE_UP_OXYGEN_AND   | 0.36837516  | 6         | 1.778e-03 | 3.105e-01 | CA2:140 C4A:195 SLC4A1:1012 RHAG:1018 CA1:2187 AQP1:8068               |
| DEURIG_T_CELL_PROLIFEROHYCTIC_LEUKEMIA_DN  | 0.05567390  | 266       | 1.818e-03 | 3.105e-01 | MAN2A1:13 TASOR:14 ABCB1:15 ZNF292:40 PTPRC:47 PAJ2:71                 |
| KINSEY_TARGETS_OF_EWSR1_FUJI_FUSION_UP     | 0.02751912  | 1172      | 1.699e-03 | 3.105e-01 | PROM1:1 KIF14:54 ORC1:65 RADX:81 WDR76:134 DDX21:145                   |
| KEGG_STEROID_HORMONE_BIOSYNTHESIS          | 0.17034477  | 28        | 1.811e-03 | 3.105e-01 | HSD17B6:146 HSD11B1:413 AKR1D1:460 CYP17A1:593 HSD17B3:600 HSD17B2:613 |
| MEISSNER_NPC_HCP_WITH_H3K27ME3             | -0.10326135 | 76        | 1.866e-03 | 3.105e-01 | GAT5:200 BAGALN2:455 SFRP5:479 TCF15:770 GRM7:827 HCK:906              |
| NIKOLSKY_BREAST_CANCER_16P13_AMPLICON      | -0.08996657 | 99        | 1.988e-03 | 3.225e-01 | C1QTNF8:43 EME2:286 NPW:340 UNKL:605 SYNGR3:738 NME3:781               |

## DisGeNET Top pathways by non-permutation

| Geneset                                  | stat        | num.genes | pval      | p.adj     | gene.vals                                                       |
|------------------------------------------|-------------|-----------|-----------|-----------|-----------------------------------------------------------------|
| Hypocalcemia                             | 0.20366287  | 45        | 2.311e-06 | 2.267e-02 | HIRA:7 CASR:63 IFT122:154 UBR1:214 DGCRC6:343 CLDN16:423        |
| DiGeorge Syndrome                        | 0.15568988  | 61        | 2.646e-05 | 9.254e-02 | HIRA:7 SPECCL1:138 ALB:142 PRODH:234 TJP1:282 DGCRC6:343        |
| Nephrocalcinosis                         | 0.16482634  | 54        | 2.830e-05 | 9.254e-02 | CLCNKB:41 NSD1:42 CASR:63 SLC26A3:273 CLDN16:423 SLC12A1:431    |
| Tetany                                   | 0.25493809  | 21        | 3.874e-05 | 9.500e-02 | HIRA:7 CASR:63 CLDN16:423 SLC12A1:431 TRPM6:624 GCM2:629        |
| Shprintzen syndrome                      | 0.20149727  | 33        | 6.218e-05 | 1.220e-01 | HIRA:7 PRODH:234 DGCRC6:343 DGCRC6L:366 CHRNA7:511 DGCRC2:538   |
| Specific learning disability             | 0.12207327  | 86        | 9.300e-05 | 1.521e-01 | HIRA:7 SPG11:25 NSD1:42 SLL1:165 PAKM:279 DGCRC6:343            |
| 22q11 partial monosomy syndrome          | 0.33633222  | 11        | 1.123e-04 | 1.574e-01 | HIRA:7 PAKA:36 JMJD1C:780 COMT:948 GNB1L:1306 ARVCF:1586        |
| Pain, Postoperative                      | 0.22719548  | 21        | 3.140e-04 | 3.850e-01 | ABCB1:15 TNF:259 TRPV1:263 SCN1LA:367 AP3B1:644 PTGCS2:7271     |
| Anemia, hereditary spherocytic hemolytic | 0.42171455  | 5         | 1.091e-03 | 6.798e-01 | ANK1:89 EPB42:568 SLC4A1:1012 SPTB:1657 SPTAL1:2807 NA          |
| Acanthosis                               | 0.30612581  | 10        | 8.023e-04 | 6.798e-01 | CAD34:66 KEL:409 VPS13A:420 XK:824 PIK3C2A:916 MTPP:1120        |
| Cervical Squamous Intraepithelial Neopla | 0.14217909  | 43        | 1.264e-03 | 6.798e-01 | APC:147 TNF:259 MML3:315 FBXW7:525 MET:558 LAMA3:617            |
| Idiopathic hypercalciuria                | 0.35501167  | 7         | 1.143e-03 | 6.798e-01 | CASR:63 SLC12A1:431 VDR:964 SLC12A3:1299 CLCN5:2330 TRPV5:2880  |
| Increased calcium level in kidney        | 0.16711728  | 33        | 8.965e-04 | 6.798e-01 | NSD1:42 CASR:63 CLDN16:423 SLC12A1:431 COL4A3:456 GCM2:629      |
| Occipital myelomeningocele               | 0.30946896  | 9         | 1.305e-03 | 6.798e-01 | HIRA:7 JMJD1C:780 COMT:948 ARVCF:1586 TBX1:2367 RNF81:2560      |
| Polyglandular Type I Autoimmune Syndrome | 0.17870421  | 30        | 7.078e-04 | 6.798e-01 | VCAM1:24 CASR:63 TNF:259 NLRP5:429 IL7:826 BPIIFB1:1083         |
| Seborrheic dermatitis                    | 0.20752574  | 20        | 1.317e-03 | 6.798e-01 | HIRA:7 NF2E2:160 TNF:259 JMJD1C:780 BTD:820 COMT:948            |
| Spherocytosis                            | 0.42171455  | 5         | 1.091e-03 | 6.798e-01 | ANK1:89 EPB42:568 SLC4A1:1012 SPTB:1657 SPTAL1:2807 NA          |
| Steroid-sensitive nephrotic syndrome     | 0.27741995  | 12        | 1.678e-04 | 6.798e-01 | TNF:259 PLA2G7:432 HPSE3:53A IL12B:1054 VEGFA:1244 GLO1:1472    |
| Unilateral primary pulmonary dysgenesis  | 0.42529439  | 5         | 9.893e-04 | 6.798e-01 | DGCRC6:343 DGCRC2:538 DGCRC8:1230 ESS2:1363 TBX1:2367 NA        |
| Congenital dyserythropoietic anemia, typ | 0.25648679  | 12        | 2.096e-03 | 7.139e-01 | MAN2A1:13 SEC23B:305 MCFD2:675 SLC4A1:1012 AICDA:1243 KLF1:1307 |
| Mental Retardation, X-Linked 1           | -0.13599417 | 42        | 2.304e-03 | 7.139e-01 | DM2:194 ARHGEF6:359 AFF2:442 ZNF71:456 CLCN4:489 MED12:1355     |
| PSEUDOHYPOALDOSTERONISM, TYPE IID        | 0.31334904  | 8         | 2.147e-03 | 7.139e-01 | NEK1:64 PRPF4B:83 WNK1:934 SLC12A3:1299 WNK4:2264               |